

Econ 1020: Principles of Micro

A plain-English study guide built from your learning guides, lectures, and syllabus.

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1 How to actually pass this class

From the syllabus and Prof. Stromme's studying slides

The grade is 80% exams, 20% problem sets, plus up to 3% extra credit from TopHat. Exams are all multiple choice, in class, closed note. Your best exam counts 30%, the other two 25% each — or you swap Midterm 3 for a comprehensive final worth 40% if it helps you.

Date	What
Sept 30 (Wed)	Midterm 1 — in class
Nov 4 (Wed)	Midterm 2 — in class
Dec 15 or 16, 3–5pm	Midterm 3 / Final — official exam slot

Two things in the syllabus quietly protect you: your lowest pset is dropped (and missing ones can be emailed at term's end for 50%), and about 5 classes of TopHat are excused. No pset extensions, ever.

The four-step method he actually recommends

- 1 · Review each slide deck right after lecture. Be ruthless: could you explain every point to a friend, out loud, with no notes? If not, you don't know it. He says 95% of students fool themselves here.
- 2 · Do the psets alone first. Every assigned problem is deliberately targeting one building block. Ask yourself: *which concept is this problem testing?* Then friend, then TA.
- 3 · Read the answer key carefully — including for questions you got right. Right answer for the wrong reason still costs you on the exam.
- 4 · Only then, practice exams. Timed, no notes. Then *modify every question*: what would make a different answer correct? Can you reverse it Jeopardy-style? Can you swap in a related concept?

The one thing he says separates A students

Memorizers do okay until a question asks them to do something *new* with a definition. The top grades go to students who learned the connections between building blocks, not just the blocks. This guide gives you the blocks — you have to build the connections by doing problems.

He also asks you to do the step-4 question-modifying yourself rather than having AI generate practice questions for you. Use this guide to learn the material fast; do the critical-thinking reps on your own.

2 Ch 1-2 · What economics is

Vocabulary chapter. Cheap points on Midterm 1 — just memorize it.

Economics is the study of how agents choose to allocate scarce resources, and how those choices affect society. An economic agent is anyone (or any group) making choices: a person, a household, a firm, a government.

	Micro	Macro
Zoom level	Individuals, households, firms, single markets	Whole economy
Example question	Why did coffee prices rise?	Why is unemployment 4%?

	Positive	Normative
What it is	Objective description, prediction, fact. Falsifiable.	What <i>should</i> happen. Moral or value judgment. Subjective.
Example	"A \$15 minimum wage reduces teen employment by 2%."	"The minimum wage should be \$15."

Exam trap

A positive statement can be *wrong* and still be positive. What makes it positive is that it *could* be checked against evidence, not that it's true.

The three principles that frame the whole course

- Optimization (Ch 3) — agents choose whatever gives them the best value.

- Equilibrium — everyone is optimizing *and* nobody believes they'd personally be better off changing their own choice, holding everyone else's choices fixed. Memorize both halves; the second half is the part students drop.
- Empiricism (Ch 2) — evidence-based analysis. Econ mostly can't run petri-dish experiments for ethical and practical reasons, so it answers cause-and-effect questions two ways: a theory-driven approach (build a model, change one thing, see what happens) and a statistics-driven approach (find clever ways to imitate a controlled experiment using observational data). Real research sits on a spectrum between them.

3 Ch 3 · Optimization & opportunity cost

The single most reused idea in the course.

Opportunity cost = the value of the best foregone alternative. In this course you use it two ways:

- As a broad idea — always ask what you're giving up.
- To put a dollar value on time — an hour spent commuting costs you what that hour was worth (usually your wage).

Sunk cost = money or time already spent that you cannot get back. *Ignore it completely* when deciding what to do next. Only future costs and benefits matter.

Cost-benefit analysis: two ways to find the same answer

Net Benefit = Total Benefit – Total Cost. The optimum is where net benefit is highest. There are two routes there:

- Total analysis — compute net benefit for every option, pick the biggest.
- Marginal analysis — look at one extra unit at a time. Keep going while marginal benefit $\geq$ marginal cost; stop when MB = MC. This is the method the rest of the course uses.

Worked example — how many hours to study, where each hour costs you \$20 of foregone work:

Hours	Total benefit	Total cost	Net benefit	MB of that hour	MC of that hour
0	\$0	\$0	\$0	—	—
1	\$60	\$20	\$40	\$60	\$20
2	\$100	\$40	\$60	\$40	\$20
3	\$125	\$60	\$65	\$25	\$20
4	\$135	\$80	\$55	\$10	\$20

Total analysis: net benefit peaks at 3 hours (\$65). Marginal analysis: hour 3 is worth taking ($25 > 20$), hour 4 isn't ($10 < 20$) — so stop at 3. Same answer, less arithmetic. That's why marginal analysis wins.

Exam trap

Marginal benefit falls as you do more of something; marginal cost usually rises. The optimum is where they cross — *not* where total benefit is highest, and *not* where cost is lowest.

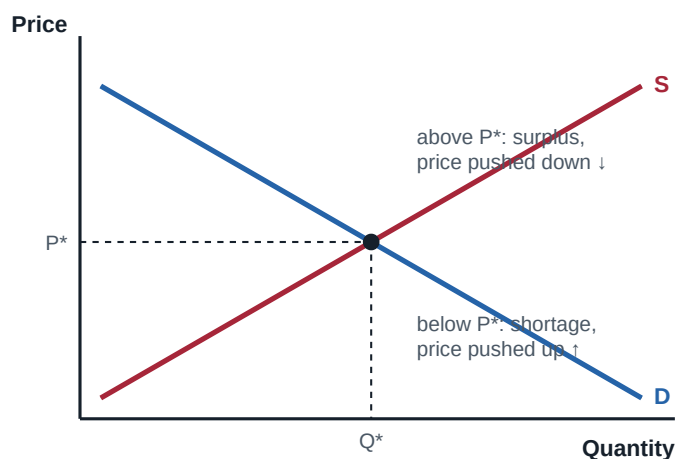
4 Ch 4 · Supply, demand, equilibrium

The workhorse model. Expect the most exam questions here.

A market is buyers and sellers of a good. *Ceteris paribus* means "holding everything else constant" — it's what lets us draw a curve in two dimensions when reality has many variables.

Reading the two curves

	Demand curve	Supply curve
Law	Price $\uparrow \rightarrow Q_d \downarrow$ (downward sloping)	Price $\uparrow \rightarrow Q_s \uparrow$ (upward sloping)
Read the height as	Willingness to pay = the buyer's marginal benefit	Willingness to accept = the seller's marginal cost
Aggregate by	Adding quantities <i>horizontally</i> at each price	Adding quantities <i>horizontally</i> at each price



Equilibrium is where the curves cross. Above P^* , $Q_s > Q_d$ — excess supply, so sellers cut price. Below P^* , $Q_d > Q_s$ — excess demand, so buyers bid price up. At P^* nobody wants to change what they're doing, which is exactly the course definition of equilibrium.

Movement along vs. shift of a curve

The #1 thing they test

A change in the good's own price = movement *along* the curve. A change in anything else = the whole curve *shifts*. If price changed on its own, do not shift the curve.

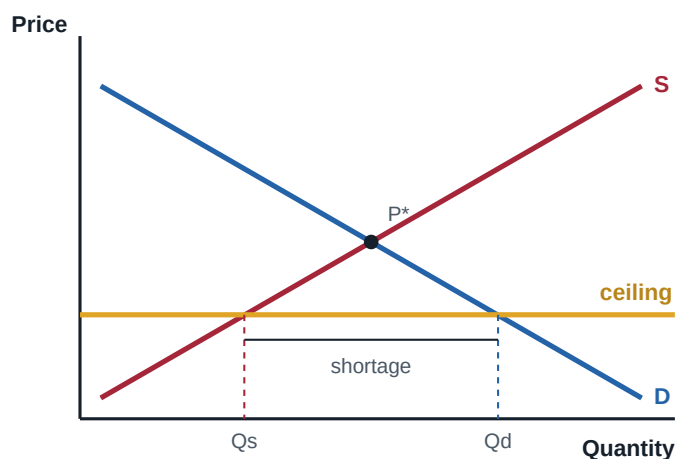
Shifts demand	Shifts supply
Income (normal vs. inferior goods) Price of substitutes (used instead — Coke/Pepsi) Price of complements (used together — printers/ink) Tastes, expectations, number of buyers	Input prices Technology Expectations Number of sellers

What happens when curves shift

Shift	Price	Quantity
Demand $\uparrow$	$\uparrow$	$\uparrow$
Demand $\downarrow$	$\downarrow$	$\downarrow$
Supply $\uparrow$	$\downarrow$	$\uparrow$
Supply $\downarrow$	$\uparrow$	$\downarrow$
D $\uparrow$ and S $\uparrow$	ambiguous	$\uparrow$
D $\uparrow$ and S $\downarrow$	$\uparrow$	ambiguous
D $\downarrow$ and S $\uparrow$	$\downarrow$	ambiguous
D $\downarrow$ and S $\downarrow$	ambiguous	$\downarrow$

Shortcut: when both curves shift, whichever effect (price or quantity) the two shifts push in the *same* direction is certain; the other one is ambiguous and depends on which shift is bigger.

Price ceilings and price floors



An effective price ceiling sits *below* equilibrium (rent control). At that low price buyers want a lot and sellers offer little, so there is excess demand — a shortage. Trade happens only up to Q_s , because sellers won't supply more. An effective price floor (minimum wage) sits *above* equilibrium and creates the mirror image: excess supply, with trade only up to Q_d .

Watch the wording

Ceiling $\rightarrow$ excess demand. Floor $\rightarrow$ excess supply. If a question's answer choices use the words "shortage" and "surplus," map them from excess demand / excess supply rather than trying to recall the label — that's where students flip the answer. A ceiling *above* equilibrium or a floor *below* it does nothing at all: it isn't binding.

5 Ch 5 · Consumers, surplus, elasticity

Elasticity is the most calculation-heavy topic on Midterm 1–2. Drill the formula.

Budget constraint

The line showing every combination of two goods you can exactly afford. With goods X and Y: $P_X \cdot X + P_Y \cdot Y = \text{Income}$.

- Income rises → line shifts out, parallel (slope unchanged).
- One price falls → line pivots outward on that good's axis only; slope changes.
- The slope is the price ratio — the rate the market lets you trade one good for the other.

Bang for your buck

Optimization at the margin says: keep spending on whichever good gives more marginal benefit *per dollar*. You're optimizing when $MB_X/P_X = MB_Y/P_Y$. If one side is bigger, buy more of that good.

Consumer surplus

CS = what you were willing to pay – what you actually paid, summed over all units bought.

From a table: subtract price from each unit's WTP and add up the positive ones. Say the price is \$6:

Unit	Your WTP	Price	Surplus
1	\$10	\$6	\$4
2	\$8	\$6	\$2
3	\$6	\$6	\$0
4	\$4	\$6	don't buy

Total CS = \$6. From a graph: CS is the area below the demand curve and above the price, out to the quantity traded. Usually a triangle, so $\frac{1}{2} \times \text{base} \times \text{height}$.

Elasticity

Elasticity answers: *by what percent does X respond when Y changes by one percent?* It's $\% \Delta X / \% \Delta Y$. Because percent changes differ depending on which point you start from, this course always uses the midpoint (arc) formula:

$$\epsilon = [(Q_2 - Q_1) / ((Q_1 + Q_2)/2)] \div [(P_2 - P_1) / ((P_1 + P_2)/2)]$$

This is the only elasticity formula you need, unless the percent changes are handed to you directly.

Example: price rises \$4 → \$6, quantity falls 100 → 80.

$\% \Delta Q = -20 / 90 = -22.2\%$. $\% \Delta P = 2 / 5 = 40\%$. $\epsilon = -22.2 / 40 = -0.56 \rightarrow$ inelastic.

$ \epsilon $	Name	Meaning	If price $\uparrow$, total revenue...
> 1	Elastic	Quantity reacts a lot	falls
$= 1$	Unit elastic	Proportional	unchanged (revenue is at its max)
< 1	Inelastic	Quantity barely reacts	rises

Rule of thumb: revenue moves in the direction of price when demand is inelastic, and against it when demand is elastic.



A straight-line demand curve has a constant *slope* but a changing *elasticity*. Elastic on the upper half, unit elastic exactly at the midpoint, inelastic on the lower half. Total revenue is largest at that midpoint.

Extreme cases: a perfectly inelastic demand curve is *vertical* ($\epsilon = 0$ — quantity never changes). A perfectly elastic one is *horizontal* ($\epsilon = \infty$ — any price increase kills all sales). Unit elastic demand is the curve where revenue stays constant.

What makes demand more elastic: more close substitutes; the good is a luxury not a necessity; it eats a large share of your budget; you're given a long time to adjust; the market is defined narrowly ("Honda Civics" is more elastic than "cars").

The other two elasticities

Elasticity	Formula	Sign tells you
Cross-price	$\% \Delta Q_d \text{ of A} \div \% \Delta P \text{ of B}$	Positive $\rightarrow$ substitutes. Negative $\rightarrow$ complements.
Income	$\% \Delta Q_d \div \% \Delta \text{income}$	Positive $\rightarrow$ normal good. Negative $\rightarrow$ inferior good.

6 Ch 6 • Producers & cost curves

Lots of formulas, but they're all just definitions. Learn the graph and the rest follows.

Vocabulary you must have cold

- Short run — at least one input is fixed. Long run — everything is variable, including whether to exist at all.

- Law of diminishing returns — adding more of one input to fixed inputs eventually raises output by smaller and smaller amounts. (Ten cooks, one oven.) This is *why* MC eventually rises.
- Economic profit subtracts *all* opportunity costs, including the salary you gave up. Accounting profit only subtracts explicit money costs. Economic profit is always $\leq$ accounting profit.
- Economies of scale — long-run average cost falls as you get bigger. Diseconomies — it rises.

Marginal product: $MP = \Delta Q / \Delta L$

$TC = TFC + TVC$

$MC = \Delta TC / \Delta Q \cdot ATC = TC / Q \cdot AVC = TVC / Q \cdot AFC = TFC / Q$

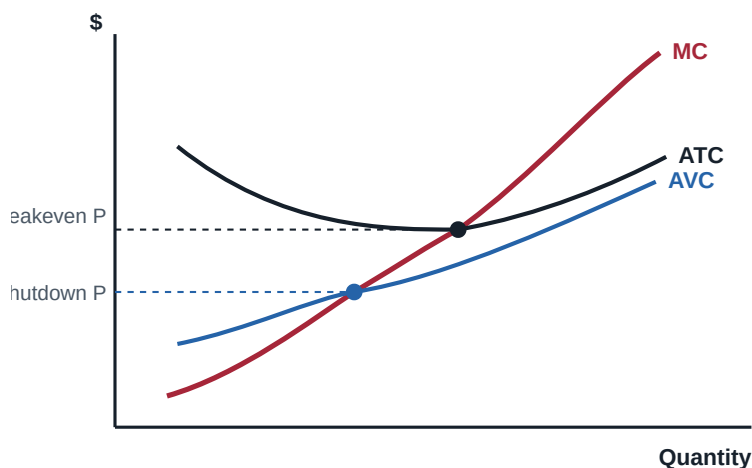
$TR = P \times Q \cdot \pi = TR - TC = Q \times (P - ATC)$

Under perfect competition: $P = MR = AR$

Exam trap he warns about directly

When computing marginals from a table, *divide by the change in Q* — don't just subtract one row from the next. If output jumps from 10 to 14 units, $MC = \Delta TC \div 4$, not ΔTC .

The cost curve picture



MC always cuts through AVC and ATC at their *minimum* points. That's not a coincidence — an average falls while the next value is below it and rises once the next value is above it. The gap between ATC and AVC is AFC, which shrinks forever as fixed cost spreads over more units.

How much to produce, and whether to produce at all

- Produce where $MR = MC$. Under perfect competition $P = MR$, so produce where $P = MC$.
- Profit or loss: compare P to ATC at that quantity. $P > ATC \rightarrow$ profit. $P < ATC \rightarrow$ loss. Size of it = $Q \times (P - ATC)$.
- Short-run shutdown: shut down if $P < \min AVC$. Above that you're at least covering variable costs and chipping away at fixed costs, which are sunk in the short run.

- Long-run exit: exit if $P < \min ATC$, since nothing is fixed anymore.
- Breakeven price = $\min ATC$ — economic profit is exactly zero there.

The firm's supply curve is its MC curve above the relevant shutdown point. Add up all firms horizontally to get market supply. In the long run, positive profits attract entry, which shifts market supply right and pushes price down — until price equals $\min$ long-run ATC and economic profit is zero. That zero-profit price is the long-run equilibrium price.

Producer surplus and supply elasticity

PS = area above the marginal cost curve, below the price, up to the quantity sold. It's the mirror image of consumer surplus.

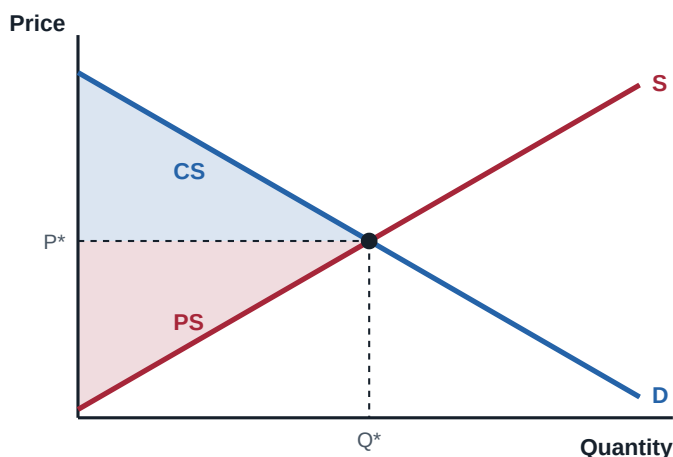
Elasticity of supply = $\% \Delta Q_S / \% \Delta P$. Ask one question: *does this make the producer more flexible?* More flexible = more elastic.

- Holding inventories (and non-perishable goods) → more elastic.
- Long run → more elastic than short run.
- Heavy reliance on fixed inputs → less elastic.
- Easy to hire/adjust variable inputs → more elastic.

7 Ch 7 · The invisible hand & efficiency

Favor the slides over the textbook here — he says so explicitly.

Decentralized decision-making means each buyer and seller chooses for themselves, guided by prices. Centralized means a planner decides. The invisible hand is the result that decentralized, self-interested choices can produce a socially good outcome — nobody intends it, and prices do the coordinating.



Total surplus = $CS + PS$ — the whole triangle between the two curves, left of Q^* . Every unit left of Q^* has marginal benefit above marginal cost, so it's worth trading. Every unit right of Q^* costs more to make than it's worth. A perfectly competitive market lands exactly on Q^* and therefore maximizes total surplus. That's what "efficient" means here.

Deadweight loss (DWL) is the surplus destroyed when a market *doesn't* trade the efficient quantity — the gap between what the market *would* do and what it *should* do. On a graph it's the triangle between the demand and supply curves over the units that failed to trade (or shouldn't have traded). Area = $\frac{1}{2} \times \text{base} \times \text{height}$, where the base is the vertical gap between MB and MC and the height is the quantity distortion.

- Pareto efficient — you cannot make anyone better off without making someone else worse off.
- Pareto improvement — a change that makes at least one person better off and nobody worse off.
- Equity–efficiency tradeoff — the biggest pie isn't necessarily fairly sliced. Efficiency is about the size of total surplus; equity is about who gets it. Policies often trade one for the other, and economics alone can't tell you the right balance — that's a normative question.

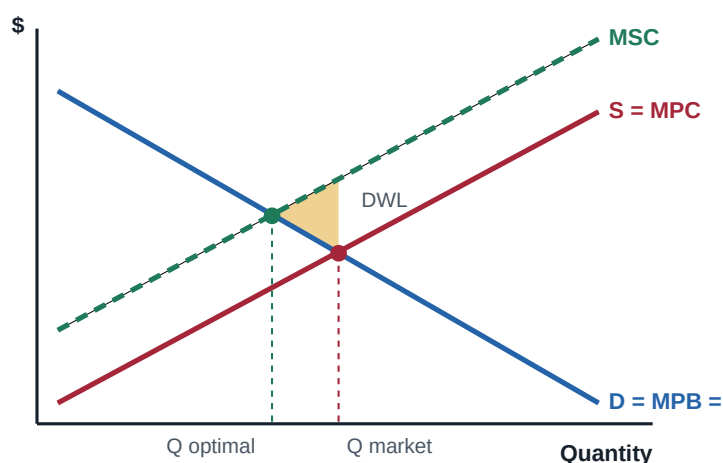
7.5 Ch 9 • Externalities

Chapter 8 is skipped — it belongs to macro.

An externality is a spillover: a cost or benefit landing on someone who wasn't part of the transaction. The market ignores it, so the market gets the quantity wrong.

Curve	What it captures
Marginal private benefit (MPB)	Value to the buyer — <i>this is the demand curve</i>
Marginal social benefit (MSB)	MPB + benefits spilling onto everyone else
Marginal private cost (MPC)	Cost to the seller — <i>this is the supply curve</i>
Marginal social cost (MSC)	MPC + costs spilling onto everyone else

Private parties trade where $MPB = MPC$. Society's optimum is where $MSB = MSC$. When those differ, you get DWL.



Negative externality (pollution): true costs exceed private costs, so MSC sits above MPC and the market overproduces. The shaded triangle is the deadweight loss — units produced whose social cost exceeded their social benefit. Flip it for a positive externality (vaccines, education): MSB sits above MPB and the market underproduces.

Fixing it

Private solutions

- Coase theorem — if property rights are clearly assigned and transaction costs are low, the parties can bargain their way to the efficient quantity *on their own*, no matter who initially holds the right. Who holds the right changes who pays whom; it doesn't change the efficient quantity.
- Coase fails when there are too many parties, rights are murky, or bargaining is expensive — which is why it doesn't solve climate change.
- "Doing the right thing" — norms and conscience internalize some externalities without any policy.

Government solutions

- Command-and-control — direct rules: caps, bans, mandated technology.
- Market-based — taxes, subsidies, tradable permits. They change prices so private incentives line up with social ones, then let firms decide how to respond. You just need to recognize why these count as market-based.

8 Ch 12 · Monopoly

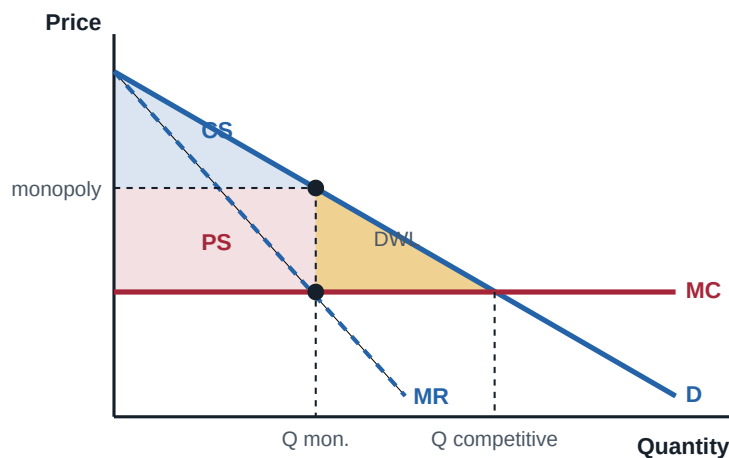
Chapters 10 and 11 are skipped.

Market power = the ability to set price above marginal cost. It's a *spectrum*, not a switch — perfect competition at one end (zero power, firms are price takers), pure monopoly at the other. Two things push a firm along it: how unique its good is, and how strong the barriers to entry are.

- Legal barriers — patents, licenses, government-granted franchises.
- Natural barriers — huge economies of scale, control of a key resource, network effects.

Why the monopolist's MR curve lies below demand

A monopolist is the market, so to sell one more unit it must lower the price — and not just on that unit, on *every* unit. So marginal revenue = the price of the new unit *minus* the revenue lost on all the previous ones. For a straight-line demand curve, MR has the same intercept and twice the slope.



Two-step move: find quantity where $MR = MC$, then go *straight up to the demand curve* to read the price. Compared to competition (where $P = MC$), the monopolist produces less and charges more. CS shrinks, part of it transfers to the firm as PS, and the rest becomes deadweight loss — mutually beneficial trades that simply never happen.

The classic monopoly mistake

Setting price where $MR = MC$. The intersection gives you the quantity only. Price always comes off the demand curve — it's what buyers will pay for that quantity.

Fixing monopoly's deadweight loss

- Private: price discrimination. You only need perfect (first-degree) price discrimination — charging every buyer exactly their willingness to pay. The result is striking: the firm produces the *efficient* quantity, so $DWL = 0$ and the outcome is efficient — but $CS = 0$ and the firm captures the entire surplus as PS. Efficient, but not equitable. A great question for testing whether you know $efficiency \neq fairness$.
- Policy: trust busting (break the firm up or block mergers) and price controls (regulate the price down toward MC).

9 Ch 13 • Game theory

Strategic situations: your best move depends on what someone else does.

Every game has three ingredients: players, strategies, and payoffs. You cover two types.

Type 1 • Simultaneous-move games (payoff matrix)

Both players move at once, or without seeing the other's choice.

- Best response — your best strategy *given* a specific choice by the other player.
- Nash equilibrium — a pair of strategies where *both* players are playing a best response, so neither would want to unilaterally switch. Notice this is just the course's definition of equilibrium applied to games.
- Dominant strategy — a strategy that's your best response no matter what the other player does. (Don't worry about the phrase "dominant strategy equilibrium" — he says it's just a Nash equilibrium with a fancier name.)

How to solve one: take Player 1's payoffs and, for each of Player 2's columns, mark P1's best. Then take P2's payoffs and, for each of P1's rows, mark P2's best. Any cell where *both* are marked is a Nash equilibrium.

		Player 2	
Player 1		Quiet	Fink
	Quiet	2 , 2	0 , <u>3</u>
	Fink	<u>3</u> , 0	<u>1</u> , <u>1</u>

Prisoner's Dilemma. Payoffs are (Player 1, Player 2); best responses underlined.

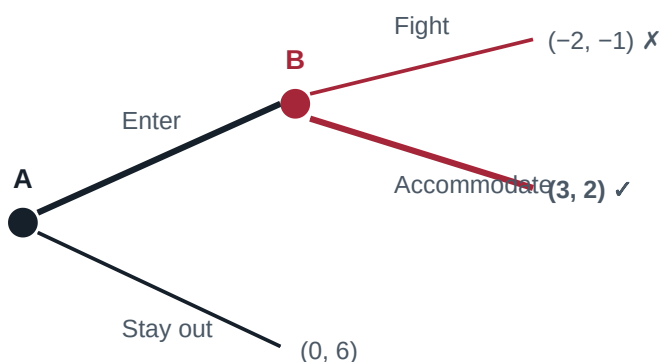
Check it: if P2 plays Quiet, P1 gets 2 from Quiet vs. 0 from Fink → Quiet. If P2 plays Fink, P1 gets 1 from Fink vs. 3 from Quiet → Quiet. So Quiet is dominant for P1, and by symmetry for P2. (Quiet, Quiet) is the unique Nash equilibrium — even though both would be better off at (Quiet, Quiet). That gap between individual rationality and the group outcome is the whole point of the dilemma.

Two things to know

Some games have one Nash equilibrium, some have several, some have none in the sense you've defined. When there's none, players mix randomly between strategies — like the penalty-kick example. That's all you need on mixed strategies. A Nash equilibrium can also be read as a social norm: a self-enforcing convention nobody wants to break alone.

Type 2 · Extensive-form games (game tree)

Players move in sequence, so the second mover sees what the first did. Solve by backward induction: start at the *end* of the tree, figure out what the last player does at each node, cross off the branches they'd never take, then work backward knowing how they'll respond.



Start at the end: if A enters, B compares Fight (−1) to Accommodate (2) and accommodates. So A's real choice is Enter → 3, or Stay out → 0. A enters. The solution is (Enter, Accommodate). B's threat to fight isn't credible, because carrying it out would hurt B.

You'll also see two games from later chapters used as examples: the fairness game and the trust and revenge game — both show real people making choices that pure self-interest doesn't predict.

10 Closing topics · Competition & information

Textbook Ch 14 and 16, but only the slide-level ideas. Ignore the rest of those chapters.

How much competition is enough?

- Results along the market-power spectrum interpolate — a market between competition and monopoly gets an outcome between the two: some markup, some DWL, but less of each than pure monopoly.
- Collusion — rival firms cooperating to act like a monopolist and keep prices high. Notice it's a prisoner's dilemma: each firm's private incentive is to cheat and undercut, which is exactly why cartels are unstable.
- You don't need many firms to get healthy competition. A handful of genuine rivals goes a long way.
- The number of firms is not very informative about concentration — market share is. Twenty firms where one holds 95% is not a competitive market.
- HHI (Herfindahl–Hirschman Index) = sum of each firm's market share squared, using percentages. Four firms at 25% each $\rightarrow 4 \times 625 = 2,500$. One firm at 100% $\rightarrow 10,000$. Higher = more concentrated. It captures share, which is why regulators use it instead of a firm count.

Asymmetric information

Asymmetric information = one side of a transaction knows something the other doesn't. It creates two distinct problems, and the exam will want you to tell them apart.

	Adverse selection	Moral hazard
Hidden what?	Hidden type	Hidden action
When?	<i>Before</i> the deal — who signs up	<i>After</i> the deal — how they behave
Example	Only sick people buy the insurance	Insured drivers drive more recklessly
Fix	Fix the information asymmetry	Fix the information asymmetry, <i>or</i> design a contract that aligns the agent's incentives with the principal's

Death spiral: insurers raise premiums to cover a sick pool $\rightarrow$ the healthiest remaining people drop out $\rightarrow$ the pool gets sicker $\rightarrow$ premiums rise again $\rightarrow$ repeat, until the market can collapse. That's adverse selection eating itself.

Principal–agent relationship: one party (the agent) acts on behalf of another (the principal) — employee/employer, contractor/homeowner, doctor/patient. Careful here: *not every principal–agent relationship has moral hazard*. It only becomes moral hazard when the principal can't observe what the agent actually does. That distinction is exactly the kind of thing he'd write a multiple-choice question around.

11 One-page formula sheet + top traps

Elasticity (midpoint): $\varepsilon = [(Q_2 - Q_1) / ((Q_1 + Q_2) / 2)] \div [(P_2 - P_1) / ((P_1 + P_2) / 2)]$

Production: $MP = \Delta Q / \Delta L$

Costs: $TC = TFC + TVC \cdot MC = \Delta TC / \Delta Q \cdot ATC = TC / Q \cdot AVC = TVC / Q \cdot AFC = TFC / Q$

Revenue & profit: $TR = P \times Q \cdot \pi = TR - TC = Q \times (P - ATC)$

Perfect competition: $P = MR = AR$; produce where $P = MC$; LR price = min ATC

Shutdown: SR if $P < \min AVC$ · LR if $P < \min ATC$ · Breakeven $P = \min ATC$

Any firm: produce where $MR = MC$

Surplus: CS = under D, above P · PS = above MC, under P · TS = CS + PS

Triangles: area = $\frac{1}{2} \times \text{base} \times \text{height}$

Social optimum: $MSB = MSC$ · Market outcome: $MPB = MPC$

HHI: $\sum (\text{each firm's \% market share})^2$

The traps that cost the most points

- Shift vs. movement. Own price changed → move along. Anything else → shift.
- Sunk costs. Already spent, already gone. Never a reason to keep going.
- Marginals from tables. Divide by ΔQ , don't just subtract rows.
- Monopoly price. Quantity from $MR = MC$, price from the demand curve above it.
- Elasticity isn't slope. A straight demand curve has one slope and many elasticities.
- Economic vs. accounting profit. Zero economic profit is normal and fine in long-run competition — it means you're earning exactly what you could elsewhere.
- Efficient ≠ fair. Perfect price discrimination is the cleanest counterexample: zero DWL, zero consumer surplus.
- Positive vs. normative. "Falsifiable in principle," not "true."
- Ceiling vs. floor. Ceiling is below equilibrium → excess demand. Floor is above → excess supply. Draw the horizontal line before you answer.
- Adverse selection vs. moral hazard. Type before the deal vs. action after it.

Use this to get the building blocks down fast. Then do what he asks in step 4 yourself: work the practice exams cold, and rewrite every question until you can see what each one is really testing. That's the part that moves a B to an A.